

Precision Moller Polarimetry in Jefferson Lab Hall-A in the 12 GeV era.

APS April 2021 Meeting “Quarks to Cosmos”

D. Eric King Jr. — Syracuse University

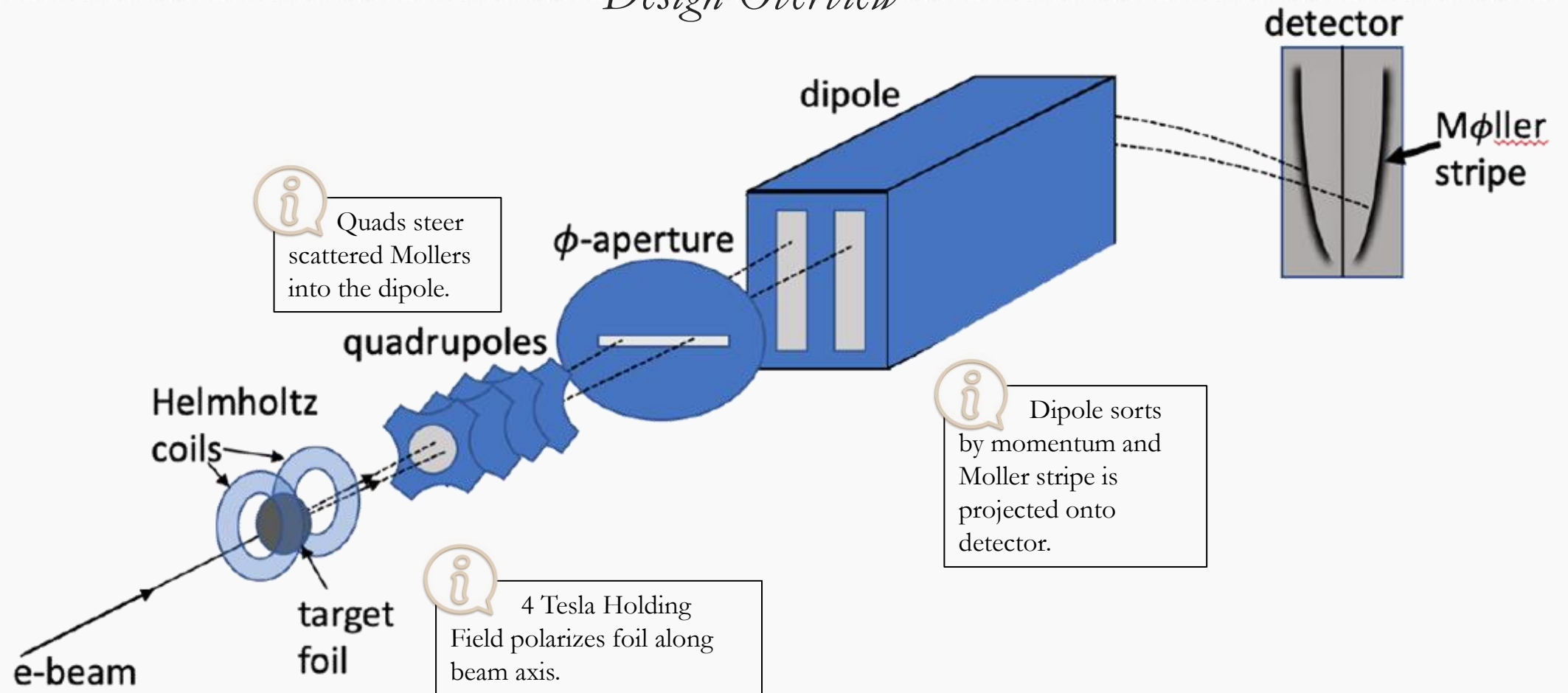


U.S. DEPARTMENT OF
ENERGY

 **Jefferson Lab**  **Syracuse University**

Hall A Moller Polarimeter

Design Overview

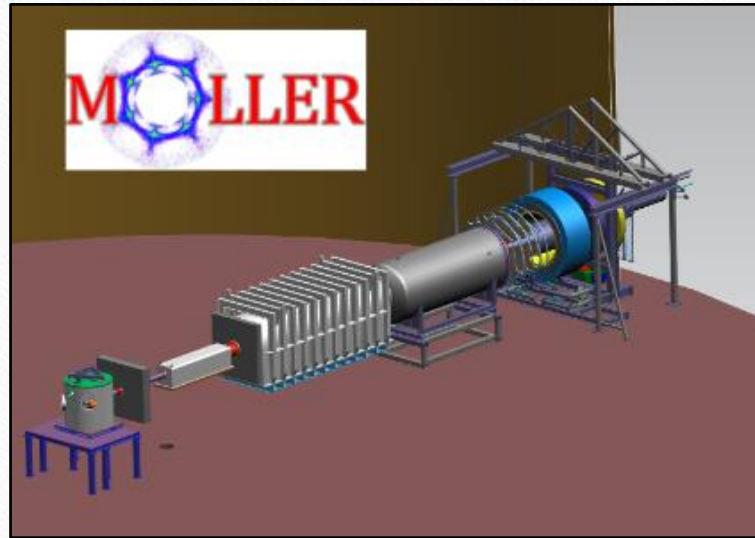


MOVING FORWARD INTO FUTURE EXPERIMENTS



PREX-II / CREX

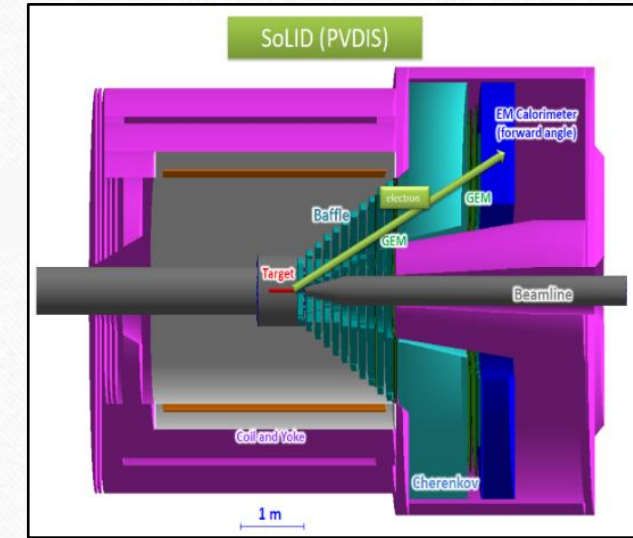
Sub-1% moller polarimetry systematics with substantial improvements in some areas. Achievements highlighted in Don Jones' previous talk.



MOLLER at 6 GeV & 11 GeV

MOLLER aims to measure A_{PV} to a precision less than one part-per-billion.

High-precision polarimetry requirement set at 0.45%.



SoLID Spectrometer

Perform ultra-precise measurements of nucleon structures; understand the emergent phenomenon of the proton mass; and many other fun and interesting things.

Polarimetry requirement is 0.40%.

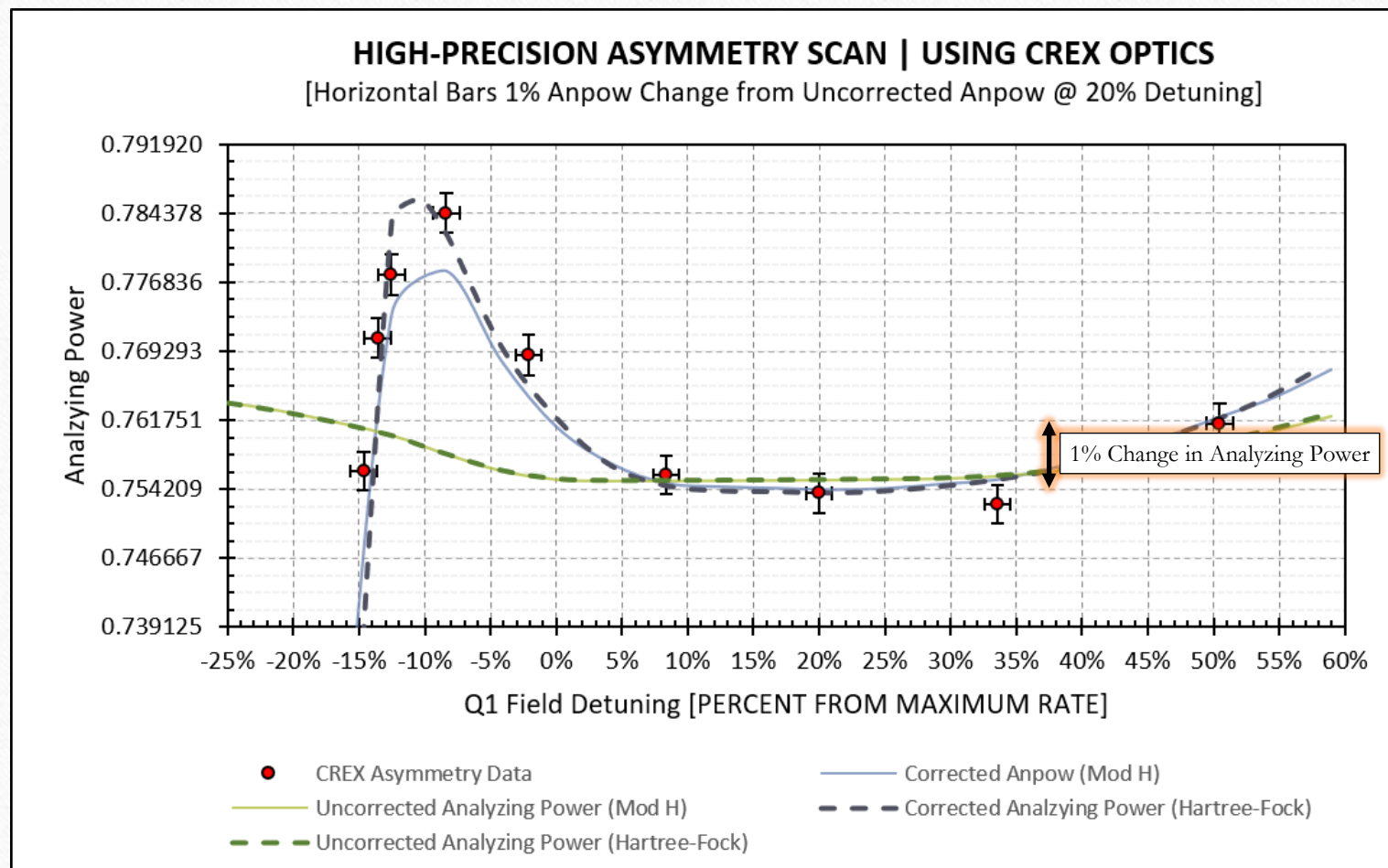
LEVCHUK EFFECT MODELING

- Proper Levchuk Effect modeling is dependent on utilizing accurate momentum distributions of Fe e-.



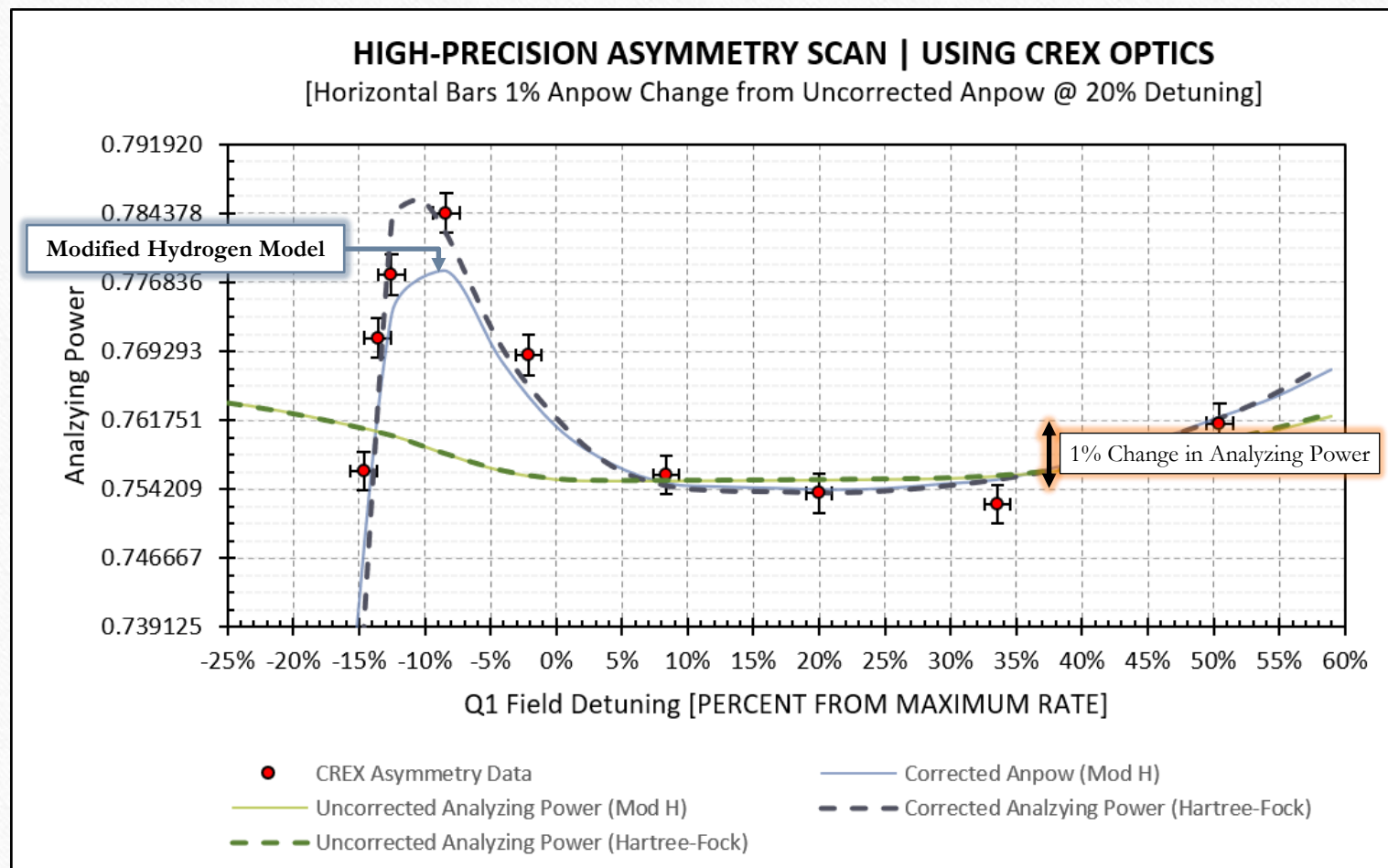
Levchuk Effect:

Møller scatters pick up a momentum kick from target electrons and in the lab frame this manifests itself as changes to the angle of scatter *but not the momentum*. This momentum kick is much larger for the tightly-bound unpolarized inner electrons compared to the loosely-bound polarized outer electrons. This can result in a dilution of unpolarized electrons reaching the detector.

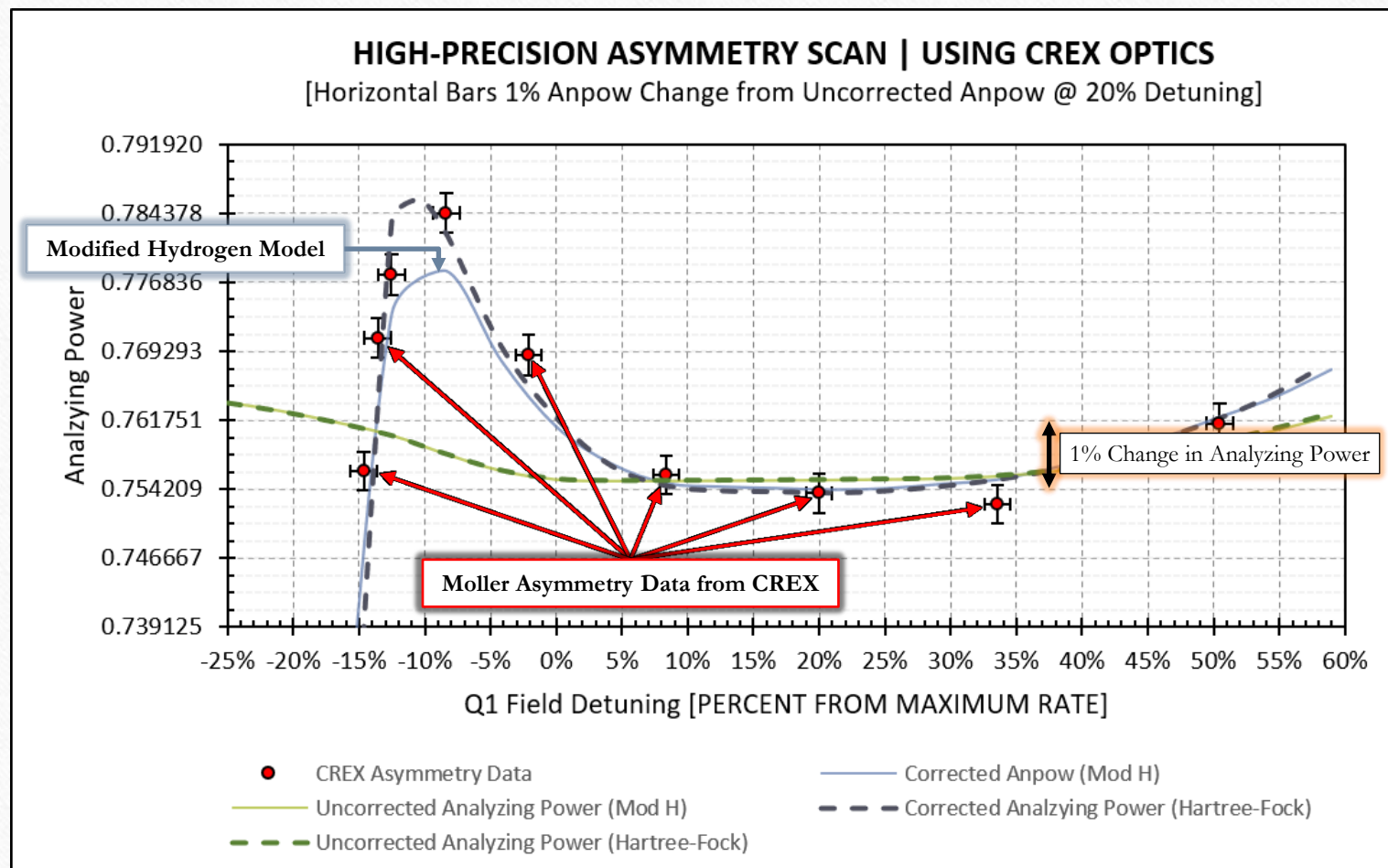


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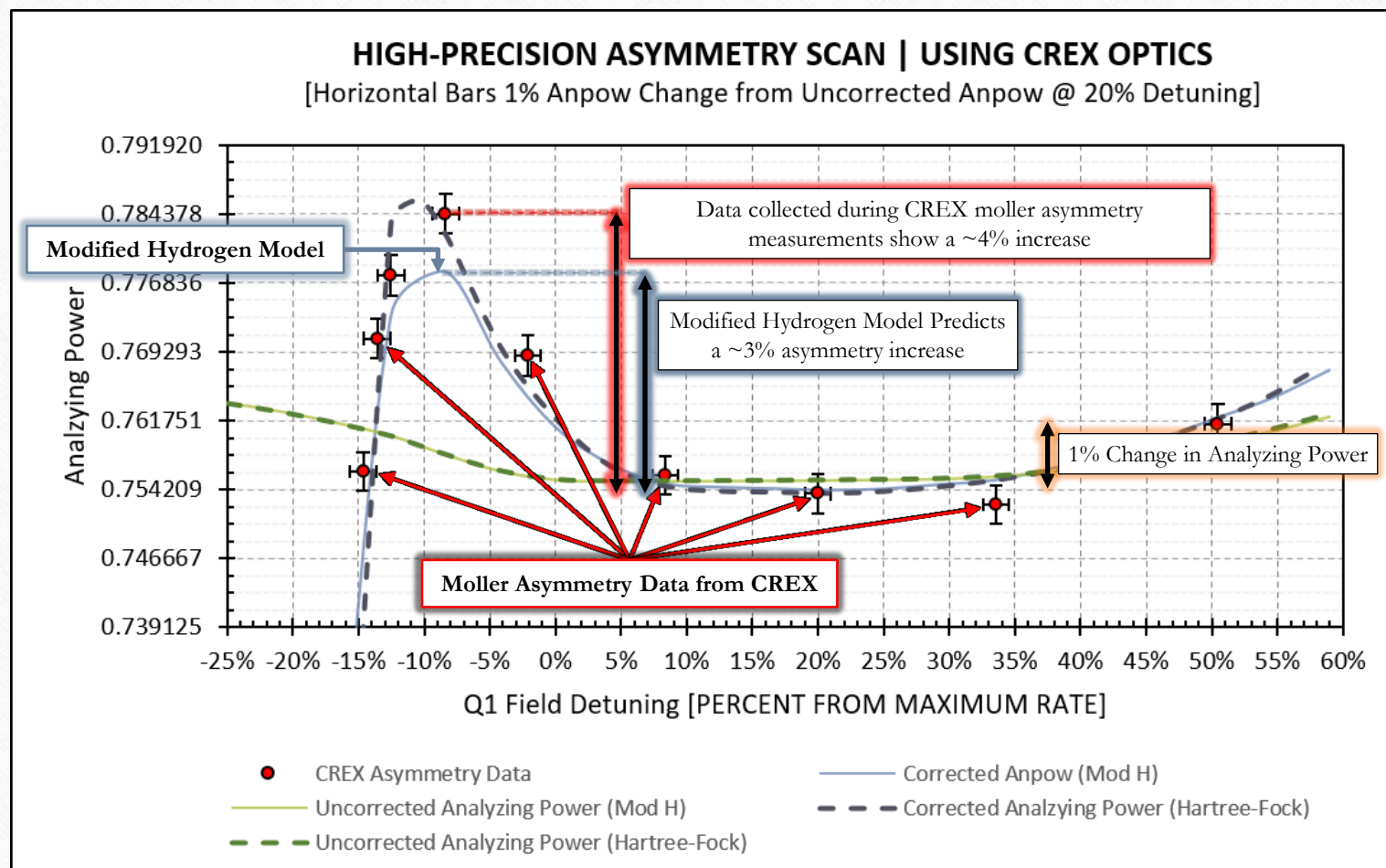


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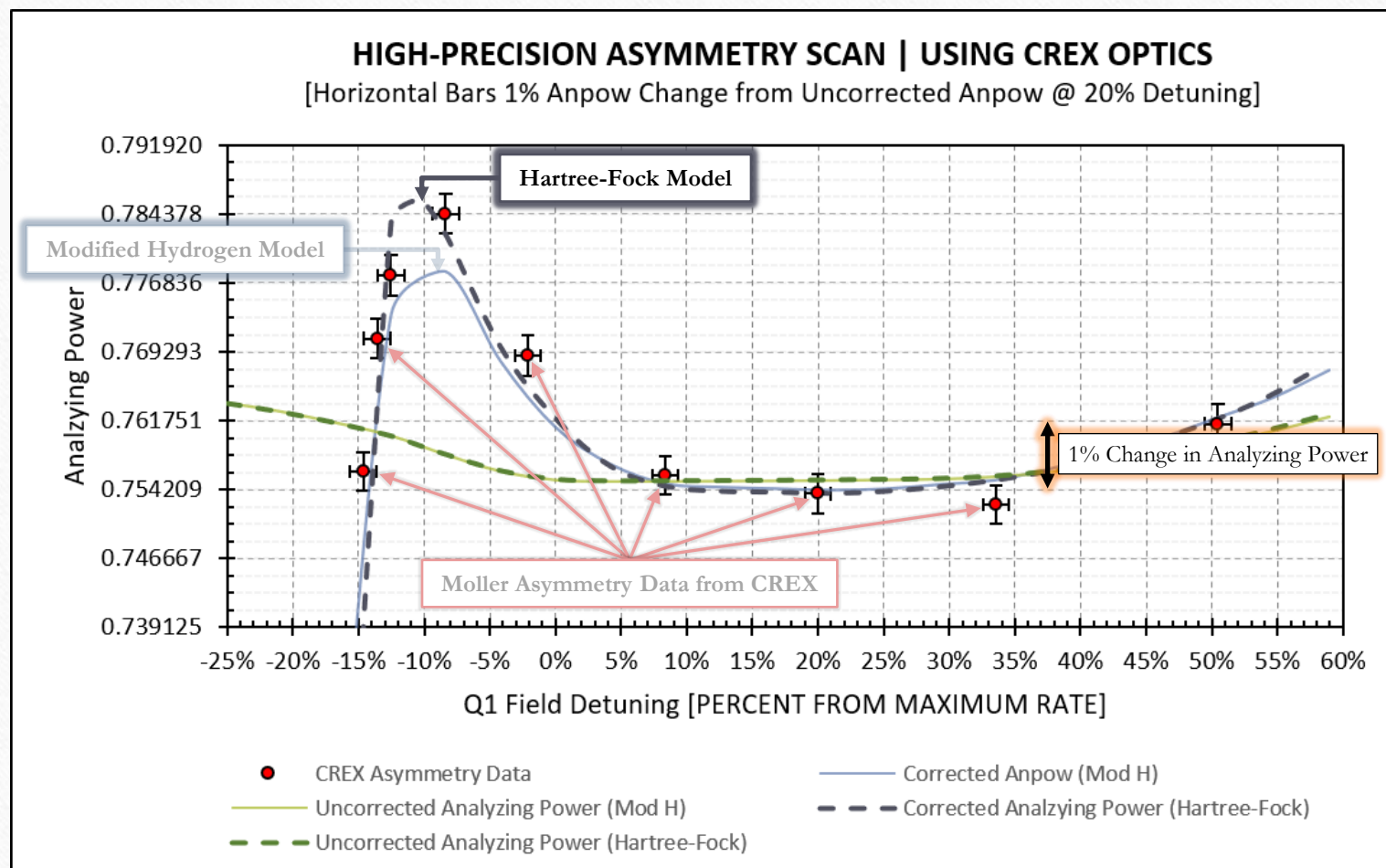
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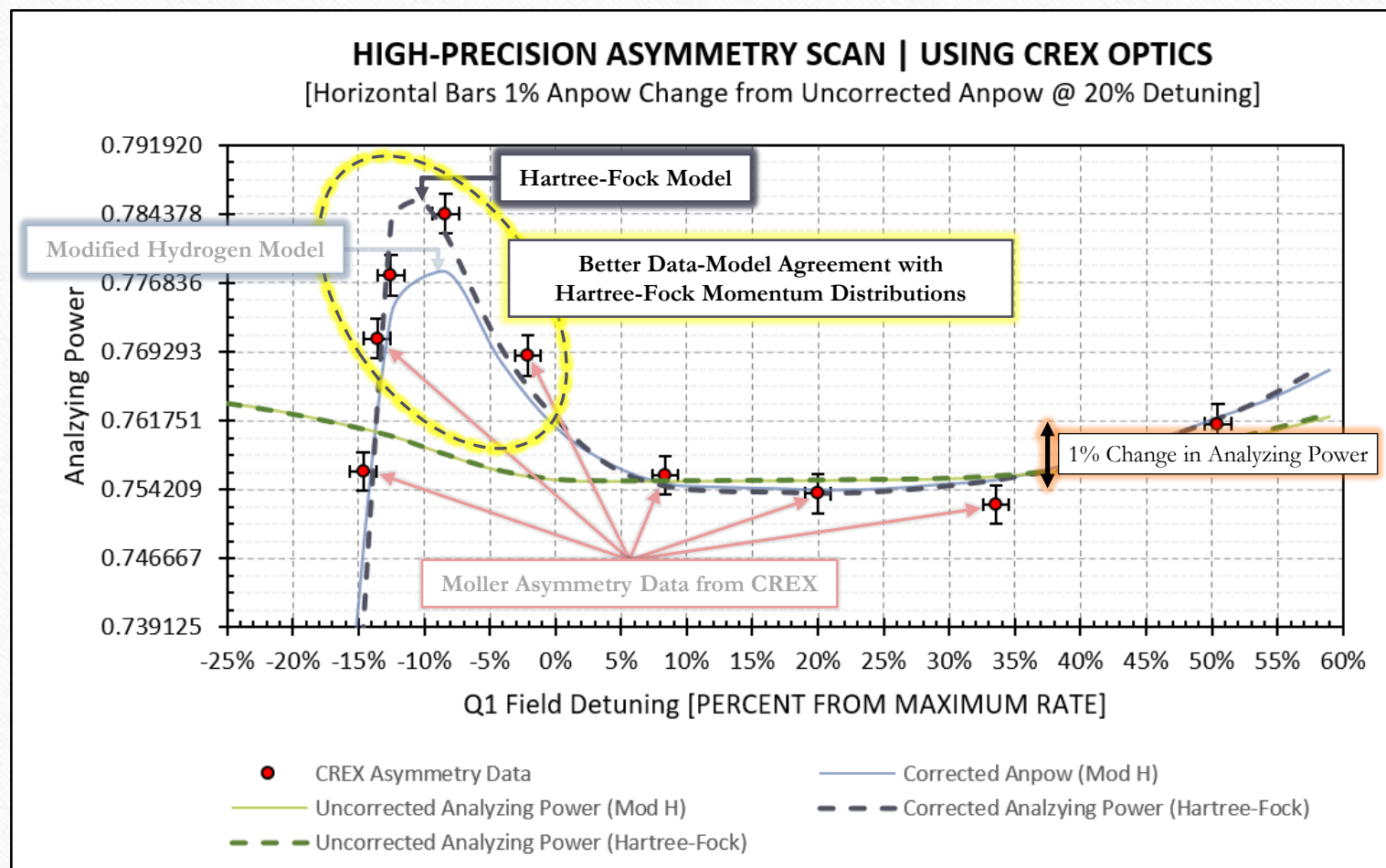
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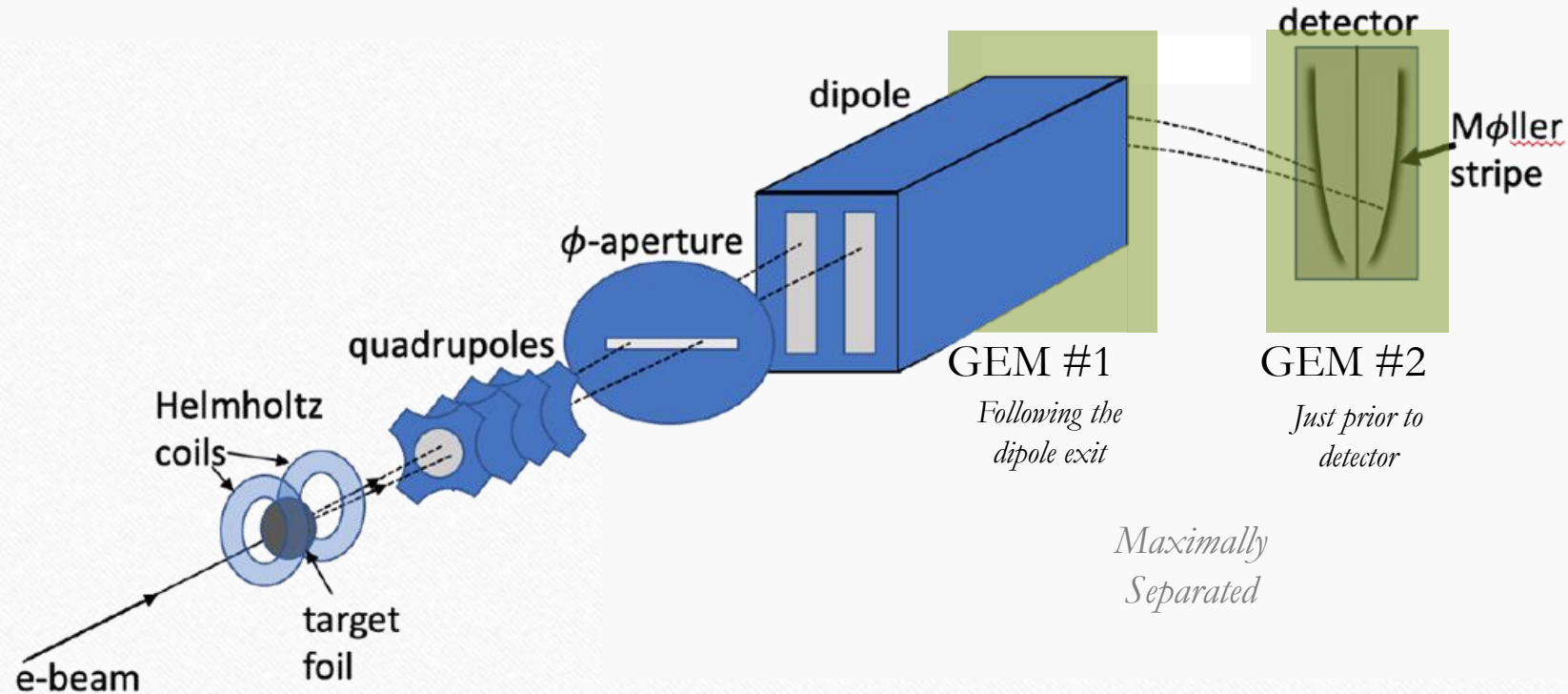
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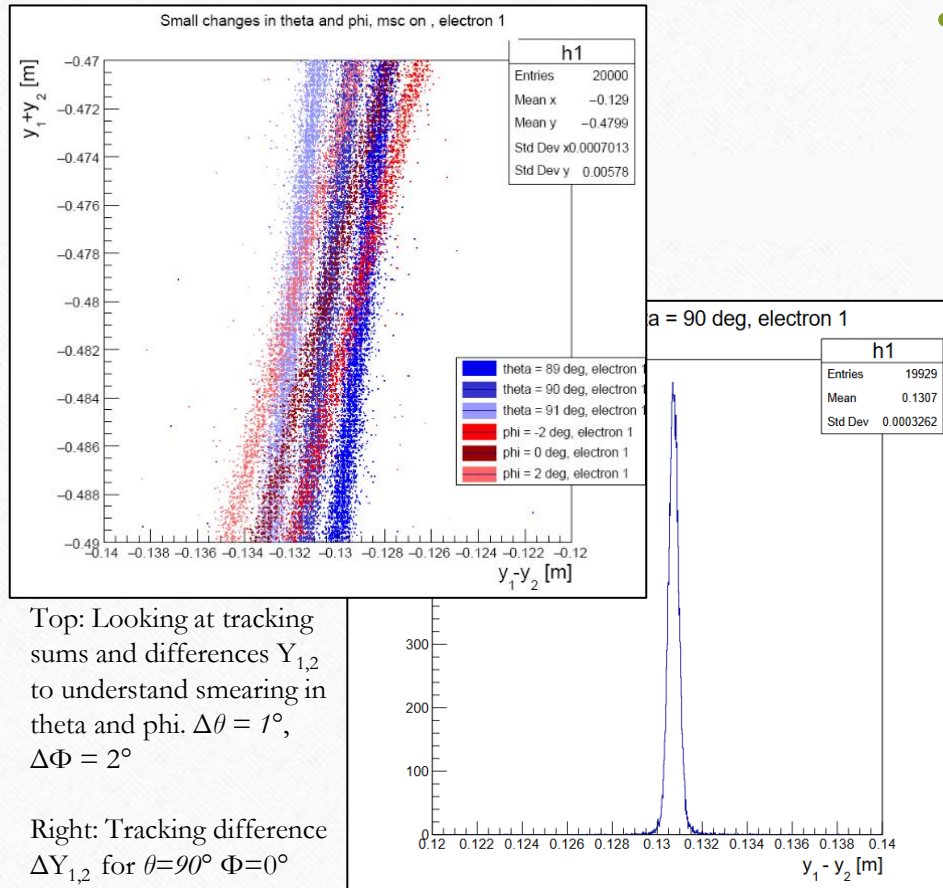
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 - Shows a ~35% difference between data and prediction at peak asymmetry [red circles].
- Utilizing Hartree-Fock model with Aaron Kaplan (Condensed Matter Graduate student at Temple University) [dashed gray line].
 - Looks to be a better data fit;
 - *Cautiously optimistic.*

POLARIMETER TRACKING

- Work has begun on understanding data which we can collect from proposed GEM tracking.
 - 2 GEM trackers are proposed.



POLARIMETER TRACKING

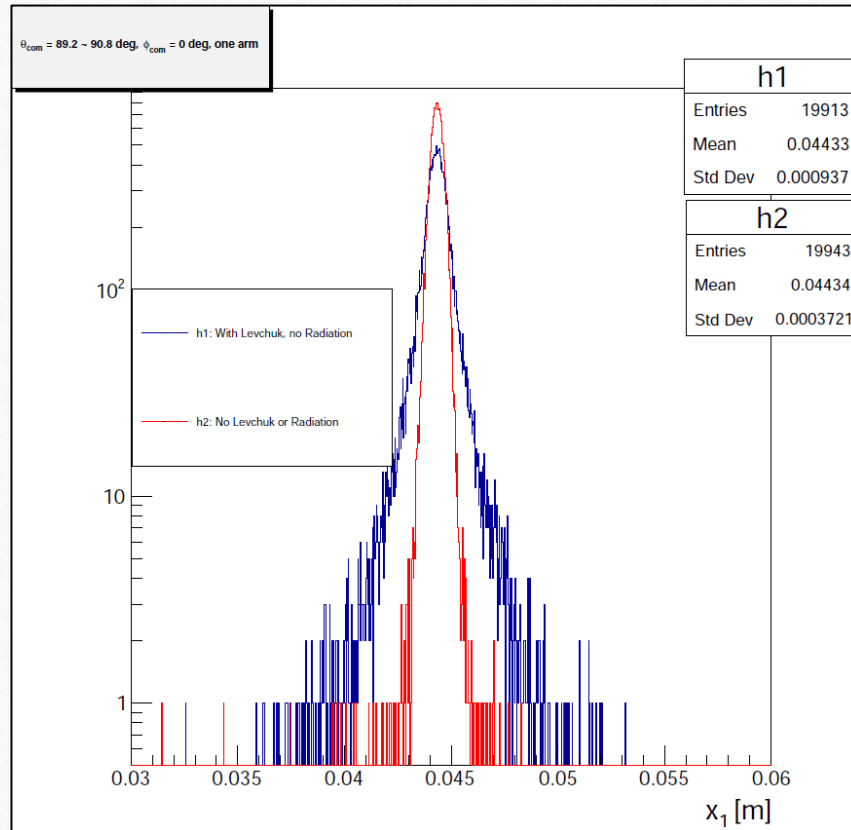


Top: Looking at tracking sums and differences $Y_{1,2}$ to understand smearing in theta and phi. $\Delta\theta = 1^\circ$, $\Delta\Phi = 2^\circ$

Right: Tracking difference $\Delta Y_{1,2}$ for $\theta=90^\circ$ $\Phi=0^\circ$ only

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 - Currently determining how much resolution will be sufficient.

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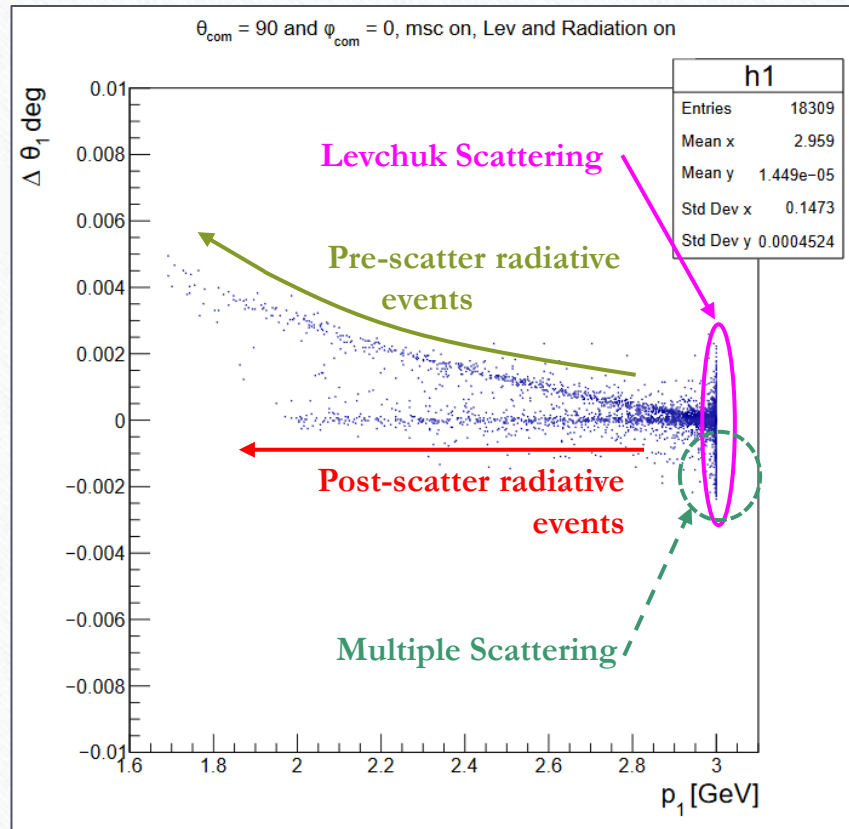


Smearing due to Levchuk Effect.

No Levchuk Effect: RED vs. **With Levchuk Effect: BLUE**

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 - Allow smearing from Levchuk to be compared to models in simulation;

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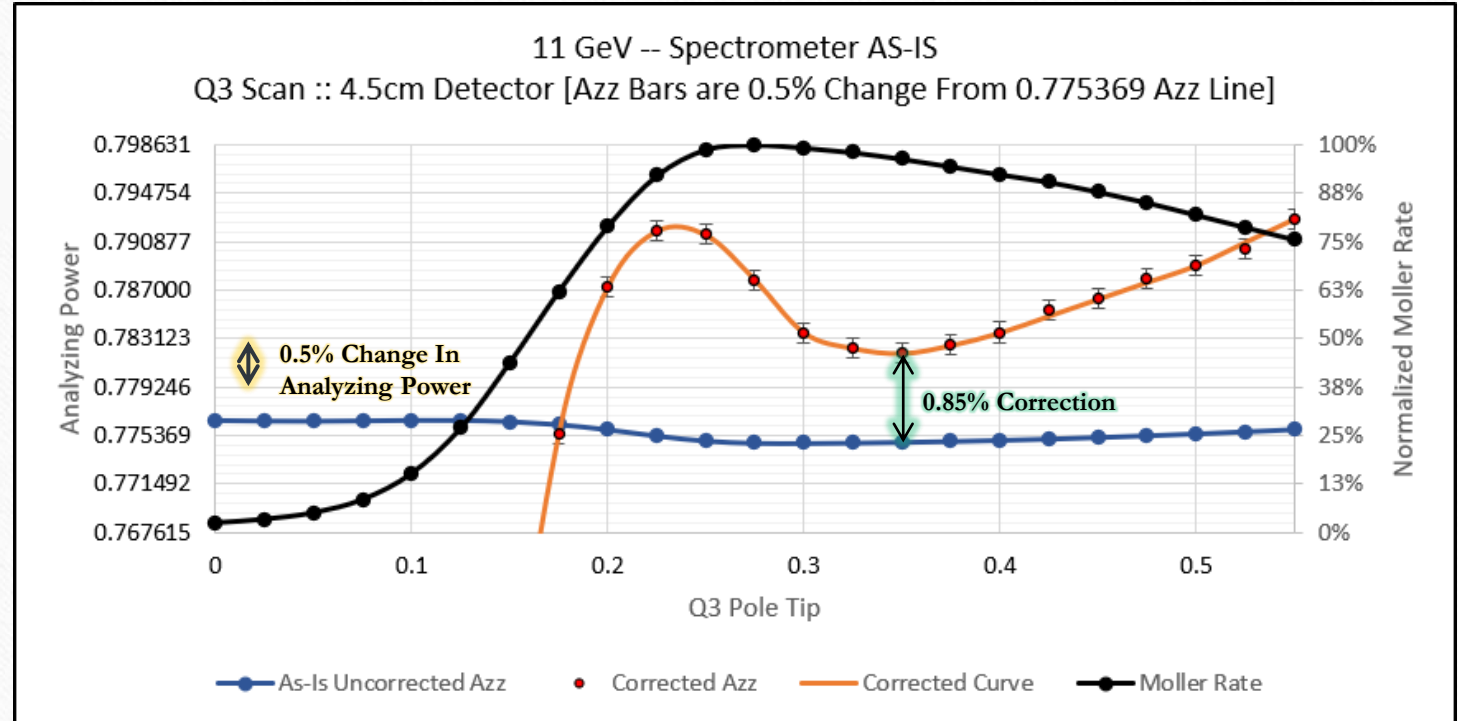


Differences in radiative effects, multiple scattering, and Levchuk events can be seen as largely separate structures in tracking data.

- Work has begun on understanding data which we can collect from proposed GEM tracking.
 - 2 GEM trackers are proposed.
 - Currently determining how much resolution will be sufficient.
- Tracking will provide a clearer understanding of Moller events.
 - Allow smearing from Levchuk to be compared to models in simulation;
 - Provide insights into radiative events—both pre- and post-scatter;
 - Provide insights on smearing effects from multiple scattering.

SUGGESTED BEAMLINE MODIFICATIONS

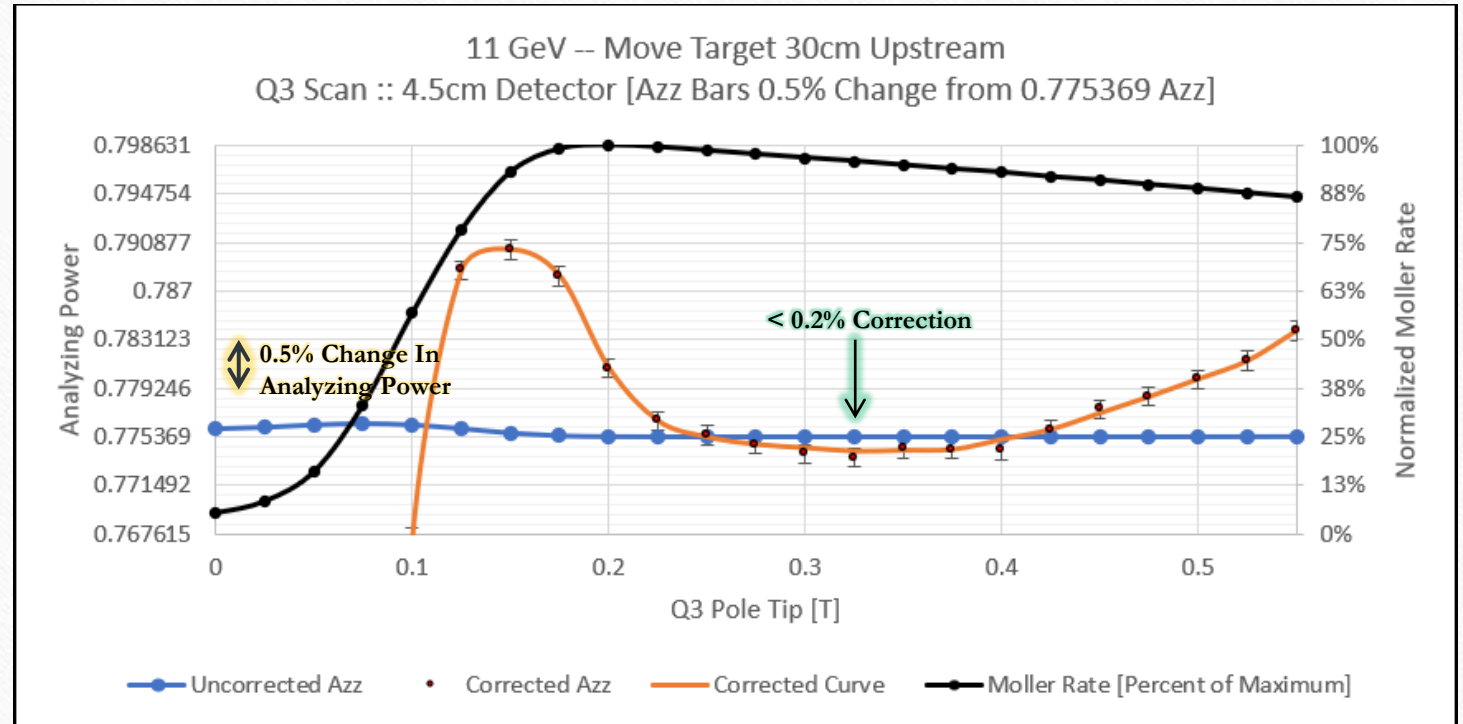
- Limiting factor in 11 GeV moller polarimetry optics is defocusing power of Quads 1 & 2.
 - As-is setup would require both to be run at maximum current in order to sufficiently guide moller scatters away from beam line.
- Our current understanding of the Levchuk effect leaves the associated systematic $\sim 0.25\%$
- In terms of our systematics budget this is unacceptable.



Analyzing power scanning on Quad3 of the spectrometer. Minimum achievable Levchuk correction is $\sim 1\%$

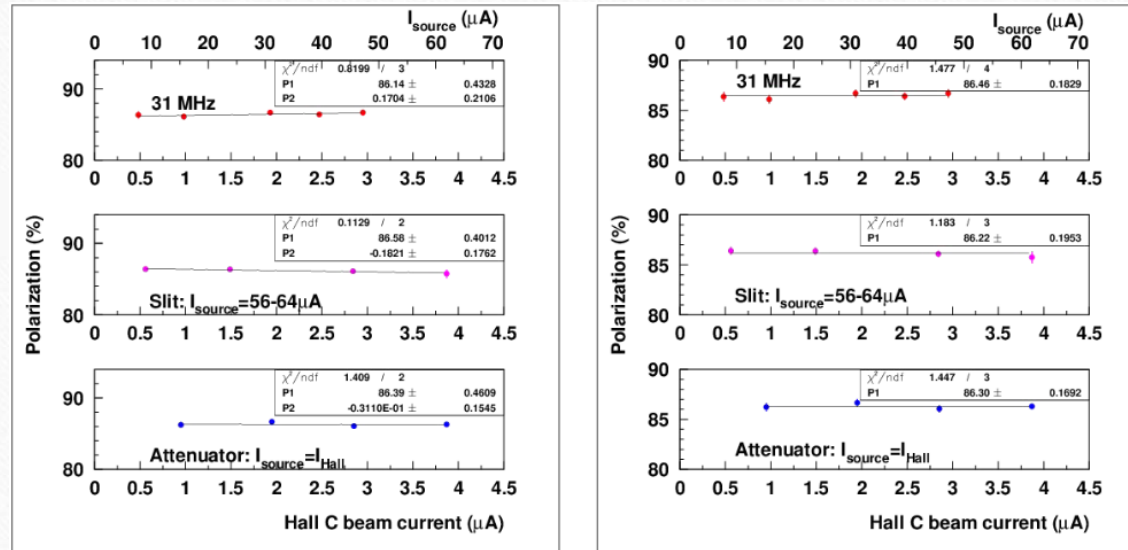
SUGGESTED BEAMLINE MODIFICATIONS

- By moving the target holding field upstream by 20-30cm we allow the moller scatters to separate, unaided by the quads, twice the amount they did previously.
 - This gives us a large region of flat analyzing power and effectively eliminates the Levchuk effect as a systematic concern
- This leaves us, as per our current understanding, with about a 0.06% systematic for corrections.
- Insights gleaned from GEM tracking data could help further constrain this systematic.



Analyzing power scanning on Quad3 of the spectrometer after moving the solenoid upstream 30cm. Correction < 0.2%

STUDY NEEDED: HIGH CURRENT EXTRAPOLATION



During Qweak Moller-Compton comparisons were performed at the same low current. At that time, Compton limited the precision to the 1% level for a $4\mu\text{A} \rightarrow 180\mu\text{A}$ dependence.

Similar tests should also be performed for MOLLER/SoLID.

- Moller polarimetry performed at $0.6\mu\text{A}$
 - Limiting factor was number of accidentals;
 - Target heating also a concern.
- MOLLER plans to run at similar current to PREX at around $70\mu\text{A}$
 - PREX systematic 0.4%.
 - It is imperative that we reduce this systematic.
- The existing study of which we are aware was performed in 2007 (shown on left) which placed limit on PREX systematic.
 - We don't have as large an error budget with MOLLER or SoLID;
 - Study needed once a worthwhile plan of investigation is decided upon.

KEY POINTS

- Improvements are currently underway in:
 - Installing tracking detectors to enhance our understanding of the polarimeter
 - Enhancements to analyzing power calculations
 - Better modeling of the Levchuk Effect
 - Physical changes to the beamline
 - Allows for control over Levchuk Effect distortions to the analyzing power.
 - Other issues such as Hall-C bleed-through, photocathode current dependence, slit dependence that can't be covered in a during the time constraints of a small talk.

Summary & Comments

Thank you

- There is a bit of remaining work, investigative and physical, which needs to be done as we move into the 12 GeV era in Hall-A at JLAB.
- We are confident, given what what's been reviewed, that we can reach the required systematic error goals for both MOLLER and SoLID.



Questions

Comments



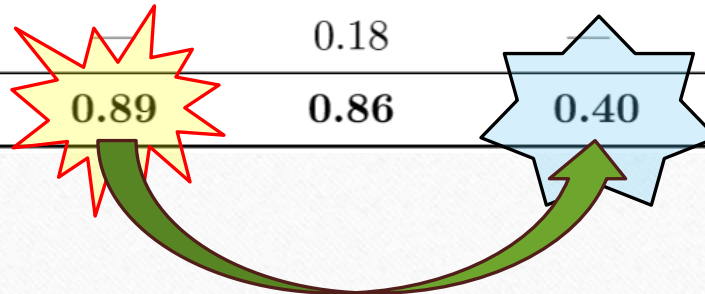
Computational work for PREX/CREX analyzing powers and 12 GeV preliminary analyzing power studies made possible by:



Syracuse University HTC Campus Grid and
NSF award ACI-1341006

IMPROVEMENTS (How we think we'll get there)

Systematic Source	PREX-II dP/P (%)	CREX dP/P (%)	Future dP/P (%)
A_{zz}	0.36	0.18	0.125
Foil Polarization	0.63	0.6	0.34
Dead Time Correction	0.05	0.135	0.05
Null Asymmetry (Cu Foil)	0.10	0.20	0.05
Accidental Correction	0.02	0.037	0.037
PITA Dependence	0.10	0.021	—
Spin Precession	0.02	0.039	0.04
Photocathode Current Dependence	0.40	0.50	0.10
Slit Dependence	0.11	0.10	0.10
July Running	0.24	—	—
Bleedthrough	—	0.18	—
Total	0.89	0.86	0.40



Projecting systematics in the future is a somewhat futile exercise; however, the road forward as it looks at the moment...

- Reduction in analyzing power uncertainties are **very feasible**.
- Don Jones' talk touched on Kerr Studies to get further our understanding of foil polarization.
- Corrections are well-understood as is what needs to be done to control them.
- Also covered by Don Jones' talk, we know what to do to control PITA dependence.
- Additional studies on photocathode current dependence will be needed.
- Hall-C bleed-through will have to be eliminated or pushed to near-zero.